

Mansi Phute

My research interests are **Responsible AI** and **ML safety**. I work on developing explanations for ML systems, analyzing them to identify vulnerabilities, and finding solutions to mitigate these issues. My UNDREAM system system offers a way to bridge differentiable rendering and photorealistic simulation for end-to-end adversarial attacks, thus enabling beter transferability of attacks to the physical world. My work includes VISOR and VISOR++ which create a universal, transferable steering image that can steer architecturally diverse models without requiring access to their internal model parameters at runtime.

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 CV PDF

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 Google Scholar

Education

Summer 2024 —	Ph.D. in Computer Science Georgia Institute of Technology, Atlanta, GA
Fall 2022 — Spring 2024	M.S. in Computer Science Georgia Institute of Technology, Atlanta, GA Specialization: Machine Learning
Fall 2018 — Spring 2022	B.Tech. in Electronics and Telecommunication Vishwakarma institute of Technology, Honors: Artificial Intelligence and Data Analytics

Research Experience

Summer 2022 — Present	Georgia Institute of Technology , Atlanta, GA <i>Graduate Research Assistant School of Computational Science and Engineering</i> Advisor: Duen Horng (Polo) Chau Member of the Polo Club of Data Science where we bridge and innovate at the intersection of data mining and human-computer interaction to synthesize scalable, interactive, and interpretable tools that amplify human's ability to understand and interact with big data. Developed defences against adversarial attacks in Language and Vision domain.
Spring 2023	Georgia Institute of Technology , Atlanta, GA <i>Graduate Teaching Assistant School of Computational Science and Engineering</i> Mentor: Duen Horng (Polo) Chau
Fall 2021 — Spring 2022	Nanyang Technological University , Singapore <i>Undergraduate Research Assistant Cyber Security Research Centre at NTU (CYSREN)</i> Mentor: Thambipillai Srikanthan Increasing python application security by analyzing libraries used. Developed dynamic dependency graph to trace vulnerabilities. Automated human resource planning and forecasting by combining business intelligence of NHS, UK with data analytics to properly shift the HR planning from manual to automated.
Spring 2021	Vishwakarma Institute of Technology , India <i>Undergraduate Research Assistant Associated with Dassault Systems</i> Mentor: Jyoti Madake Developed AI based solutions for agricultural problems faced in India by using hyperspectral imaging to predict soil fertility in the land
Fall 2020	Vishwakarma Institute of Technology , India <i>Undergraduate Research Assistant School of Electronics and Telecommunication</i> Mentor: Abha Marathe Conducted a thorough literature survey on the use of AI in finance and the various ways it is used for risk management in the stock market

Industry Research Experience

Summer 2026	IBM. , San Jose, CA <i>Research Intern AI Foundations Team</i> Mentor: Nathalie Baracaldo, Heiko Ludwig Worked on a way to enforce policy on Agent to agent communication in KV cache space
Summer 2025 — Fall 2025	HiddenLayer, Inc. , Austin, TX <i>Research Intern Adversarial Robustness Team</i> Mentor: Jason Martin, Ravi Balakrishnan Helped pioneer transition of AI defense systems to account for multimodal attacks. Developed universal transferrable multimodal steering images that can alter model behavior using the input channel without requiring access to the model internals. My work during the internship was implemented into the product AIDR (AI Detection and Response) (https://www.hiddenlayer.com/aidr/)
Summer 2019	Tech Mahindra Ltd , Pune, India <i>Intern, Web Development</i> Mentor: Rahul Bedmutha Developed a portal for internal use, using HTML, CSS and Javascript.

Honors and Awards

2024	Marshall D. Williamson Fellowship Awarded to a well-rounded, second-year Master's student who best embodies values of academic excellence and leadership
2022	Best Scholar in ECE Merit-based award for the ECE student with the highest undergraduate GPA in the entire department

Publications

ALLUDE: A Unified Evaluation System for Configurable Attacks in Differentiable Environments

Mansi Phute, Alex Greenhalgh, Matthew Hull, Haoran Wang, Alec Helbling, ShengYun Peng, Willian Lunardi, Martin Andreoni, Wenke Lee, Duen Horng (Polo) Chau
arXiv. 2026.

  

UNDREAM: Bridging Differentiable Rendering and Photorealistic Simulation for End-to-end Adversarial Attacks

Mansi Phute, Matthew Hull, Haoran Wang, Alec Helbling, ShengYun Peng, Willian Lunardi, Martin Andreoni, Wenke Lee, Duen Horng (Polo) Chau
Pacific Asia Conference on Knowledge and Data Discovery. 2026.

  

VISOR++ - Transferrable Visual Input based Steering for Output Redirection in Large Vision Language Models

Ravi Balakrishnan*, Mansi Phute*
European Conference on Computer Vision 2026 (ECCV 2026). 2026.

  

VISOR - Visual Input based Steering for Output Redirection in Large Vision Language Models

Mansi Phute, Ravi Balakrishnan
Assessing and Improving Reliability of Foundation Models in the Real World Workshop (AAAI AIR-FM Workshop). 2026.

  

ComplicitSplat: Downstream Models are Vulnerable to Blackbox Attacks by 3D Gaussian Splat Camouflages

Matthew Hull, Haoyang Yang, Pratham Mehta, **Mansi Phute**, Aeree Cho, Haoran Wang, Matthew Lau, Wenke Lee, Willian Lunardi, Martin Andreoni, Duen Horng Chau
arXiv (arXiv). 2025.

  

3D Gaussian Splat Vulnerabilities

Matthew Hull, Haoyang Yang, Pratham Mehta, **Mansi Phute**, Aeree Cho, Haoran Wang, Matthew Lau, Wenke Lee, Willian Lunardi, Martin Andreoni, Duen Horng Chau
CVPR Workshop on Neural Fields Beyond Conventional Cameras (NFBCC) (CVPR'25). 2025.

  

Interpretation Meets Safety: A Survey on Interpretation Methods and Tools for Improving LLM Safety

Seongmin Lee, Aeree Cho, Grace C. Kim, ShengYun Peng, **Mansi Phute**, Duen Horng (Polo) Chau
Main, Conference on Empirical Methods in Natural Language Processing (EMNLP). 2025.

  

RenderBender: A Survey on Adversarial Attacks Using Differentiable Rendering

Matthew Hull, Haoran Wang, Matthew Lau, Alec Helbling, **Mansi Phute**, Chao Zhang, Zsolt Kira, Willian Lunardi, Martin Andreoni, Wenke Lee, Duen Horng Chau
International Joint Conference on Artificial Intelligence (IJCAI) (IJCAI'25). 2025.

 

Semi Truths: A Large-Scale Dataset for Testing Robustness of AI-Generated Image Detectors

Anisha Pal, Julia Kruk, **Mansi Phute**, Manogna Bhattaram, Diyi Yang, Duen Horng (Polo) Chau, Judy Hoffman
NeurIPS. 2024.

   

LLM Attributor: Interactive Visual Attribution for LLM Generation

Seongmin Lee, Zijie J. Wang, Aishwarya Chakravarthy, Alec Helbling, ShengYun Peng, **Mansi Phute**, Duen Horng (Polo) Chau,, Minsuk Kahng
ACL demo. 2024.

  

Robust Principles: Architectural Design Principles for Adversarially Robust CNNs

ShengYun Peng, Weilin Xu, Cory Cornelius, Matthew Hull, Kevin Li, Rahul Duggal, **Mansi Phute**, Duen Horng (Polo) Chau, Jason Martin
BMVC. 2023.

      

LLM Self Defense: By Self Examination, LLMs Know They Are Being Tricked

Mansi Phute, Alec Helbling, Matthew Hull, ShengYun Peng, Sebastian Szyller, Cory Cornelius, Duen Horng (Polo) Chau
ICLR Tiny Paper. 2024.

    

Patent

April 2026	"US12608861B1: Model output steering using universal images
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Talks and Presentations

March 2025	Large Language Model Evaluation Georgia Institute of Technology, CS 8001: Large Language Models ★ Invited
June 2024	Large Language Models and How They Work Georgia Institute of Technology, CS 8001: Large Language Models ★ Invited
October 2023	LLM Self Defense: By Self Examinations, LLMs Know They Are Being Tricked! IBM, San Jose CA

Press

April 2024	"Student Excellence Honored at Annual Event," Georgia Tech
August 2023	"GRE Success Stories: How Test Takers Scored Above the 90th Percentile," Jamboree
May 2020	"Team Eklavya- E&TC; students team Designs Autonomous sanitisation robot," Vishwakarma Institute of Technology

Teaching

Spring 2023	Graduate Teaching Assistant <i>Georgia Institute of Technology, Atlanta, GA</i> Data and Visual Analytics (CSE 6242 / CX 4242), Instructor: Duen Horng (Polo) Chau I was a Teaching Assistant (TA) at Georgia Tech for the class Data and Visual Analytics where I worked with a team of 30 TAs to enable learning in a class of more than 1200 students. I was a part of designing homework and mentoring students in their course work and project work.
Fall 2019	Teaching Volunteer <i>Vishwakarma Institute of Technology, Pune, India</i> Aashadeep: Literacy Program for underprivileged people, Instructor: A semester long teaching program where I created learning opportunities for increasing literacy in society aimed towards people outside the traditional schooling age. Thus proving that there is no binding of age to learn how to read or write. This program aimed at combating illiteracy in specific sections of society.

Service

Reviewer NeurIPS Workshop on Socially Responsible Language Modelling (NeurIPS SoLaR) 2023 International Conference on Learning Representations (ICLR) 2026

Mentoring

Sri Ranganathan Palaniappan
B.S. in Computer Science, Georgia Institute of Technology

References

Dr. Polo Chau, Associate Professor
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Georgia Institute of Technology
cc.gatech.edu/~dchau/

Dr. Thambipillai Srikanthan, Professor
School of Computer Science and Engineering
Nanyang Institute of Technology
www.ntu.edu.sg/scse/about-us/past-chairs/prof-thambipillai-srikanthan